

Robustness or Weighting? Re-Evaluating Federated Aggregation on Heterogeneous Health Data

Kunal Chevuri and Dr. Mohammed Aledhari

Data Science Department

College of Artificial Intelligence and Advanced Analytics · University of North Texas

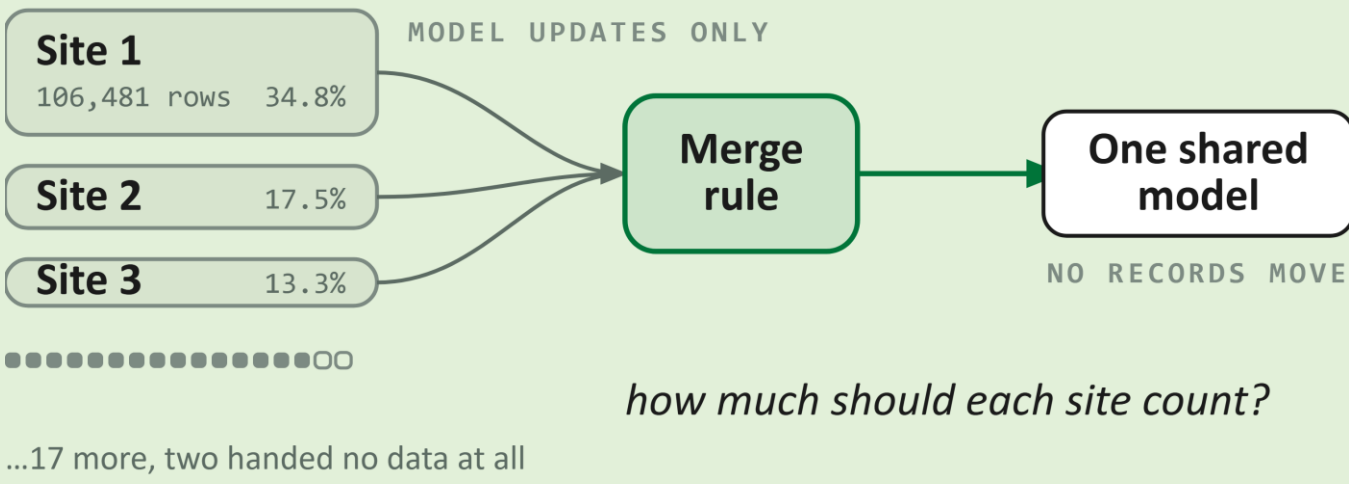


Motivation & Questions

- Hospitals and health agencies hold records that could sharpen how we predict disease, but privacy law stops them pooling the data.
- Federated learning trains one shared model without moving any patient record out of the institution holding it.
- The rule that merges each site's work is normally chosen for its robustness to noisy or unusual sites.
- Robust rules are always scored against FedAvg, which weights each site by how much data it holds, so the biggest site gets the loudest vote.
- Robust rules discard that weighting entirely, so two things change at once and nobody had separated them.
- We ask: when a robust rule wins, is it the robust statistic winning, or just the end of the loudest vote?

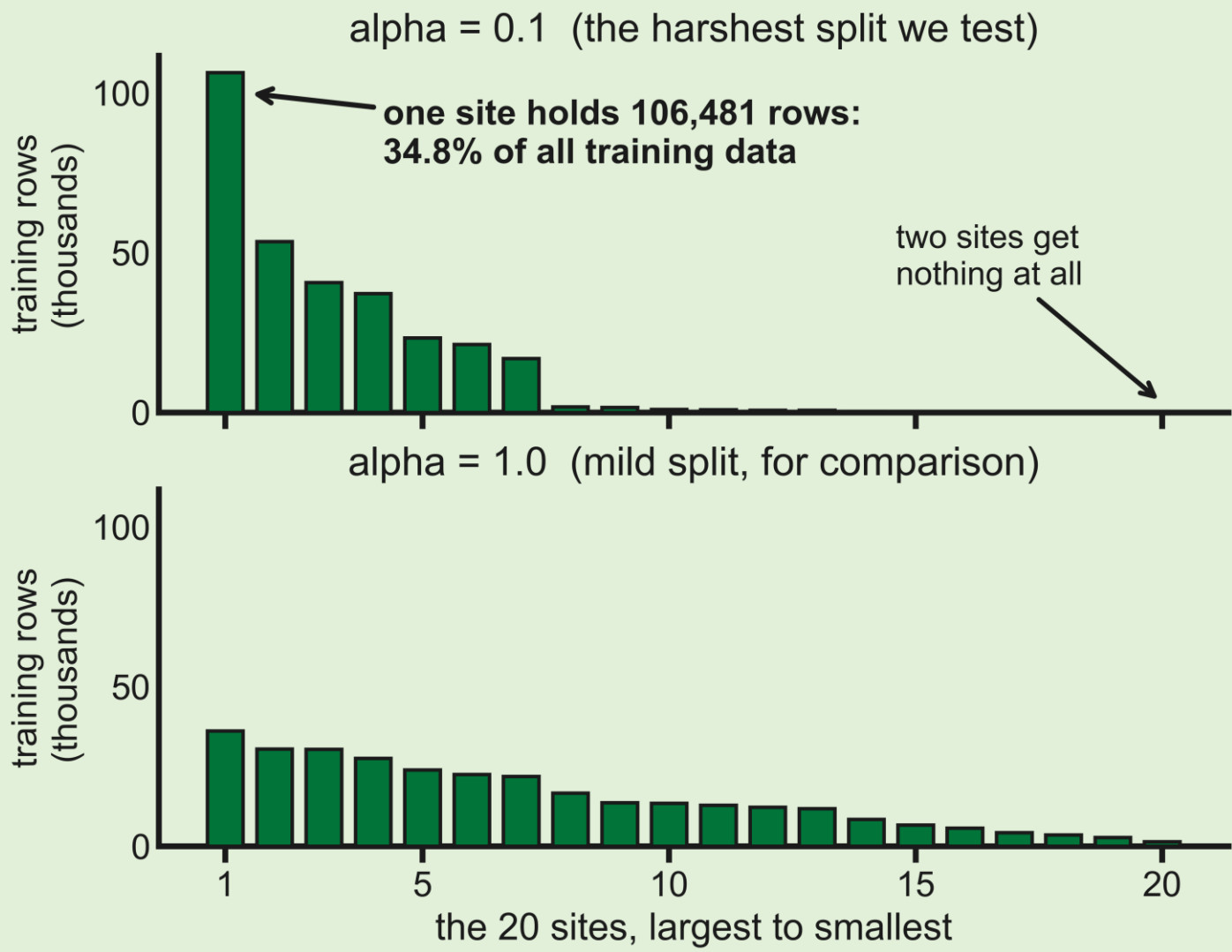
Background: how federated learning works

- Each site trains on its own data and sends only the model update, which the server merges into one shared model.



Dataset: two health datasets, split 20 ways

- CDC BRFSS 2023: 382,709 U.S. adults, predicting self-reported depression, which 20.7% of respondents report.
- UCI Breast Cancer Wisconsin runs alongside as a clean clinical benchmark.
- Each dataset is dealt to 20 simulated sites by Dirichlet partitioning, whose dial alpha sets how lopsided the split is.



- At alpha = 0.1 one site holds 106,481 training rows, 34.8% of the total, while two sites receive nothing at all.

References

[1] McMahan et al. Communication-efficient learning of deep networks from decentralized data. AISTATS, 2017.

[2] Blanchard et al. Machine learning with adversaries: Byzantine tolerant gradient descent. NIPS, 2017.

[3] Yin et al. Byzantine-robust distributed learning: towards optimal statistical rates. ICML, 2018.

[4] Karimireddy et al. Byzantine-robust learning on heterogeneous datasets via bucketing. ICLR, 2022.

[5] Hsu et al. Measuring the effects of non-identical data distribution for federated visual classification. arXiv:1909.06335, 2019.

[6] CDC. BRFSS 2023 survey data and documentation.

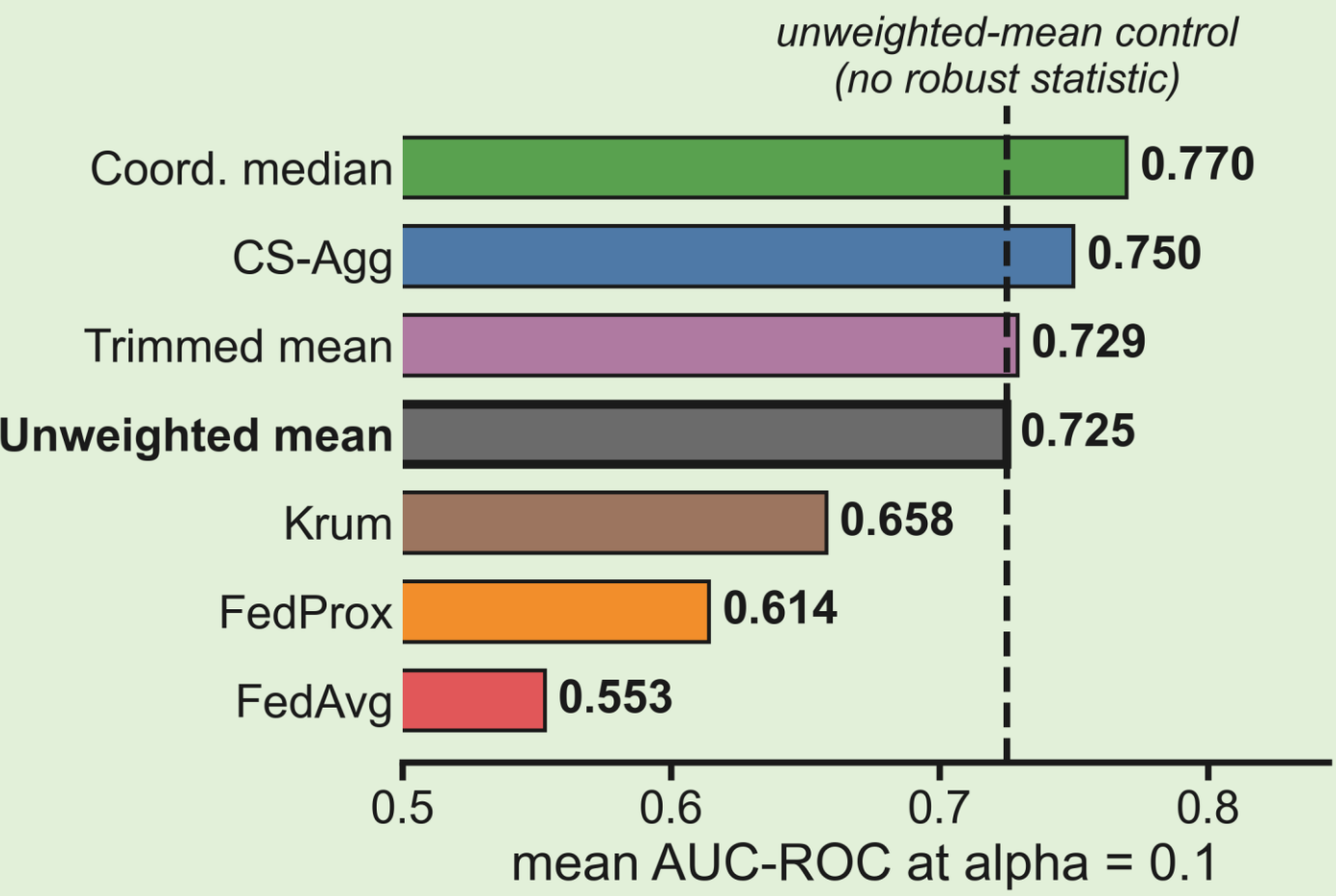
Methods: seven merge rules and one control

- Seven merge rules run head to head: FedAvg, FedProx, CS-Agg, Krum, trimmed mean, coordinate-wise median, and an unweighted mean.
- The unweighted mean is our control: it uses no robust statistic and differs from FedAvg only in ignoring how much data each site holds.
- Anything the control gains over FedAvg is therefore weighting alone, which is what isolates the effect.
- On BRFSS we corrupt labels only the way real people err, reporting no depression when they have it.

What we varied	Levels tested
Merge rules	7, including the unweighted-mean control
Datasets	BRFSS 2023 and Breast Cancer Wisconsin
Split skew, alpha	0.1, 0.5, 1.0 — lower is more lopsided
Label noise	0, 10, 20, 30% of records, at 20% of sites
Random seeds	42, 123, 456
Every run	20 sites, 50 rounds, scored by AUC-ROC
Total	624 completed runs

Results

Most of the gain over FedAvg is weighting, not robustness

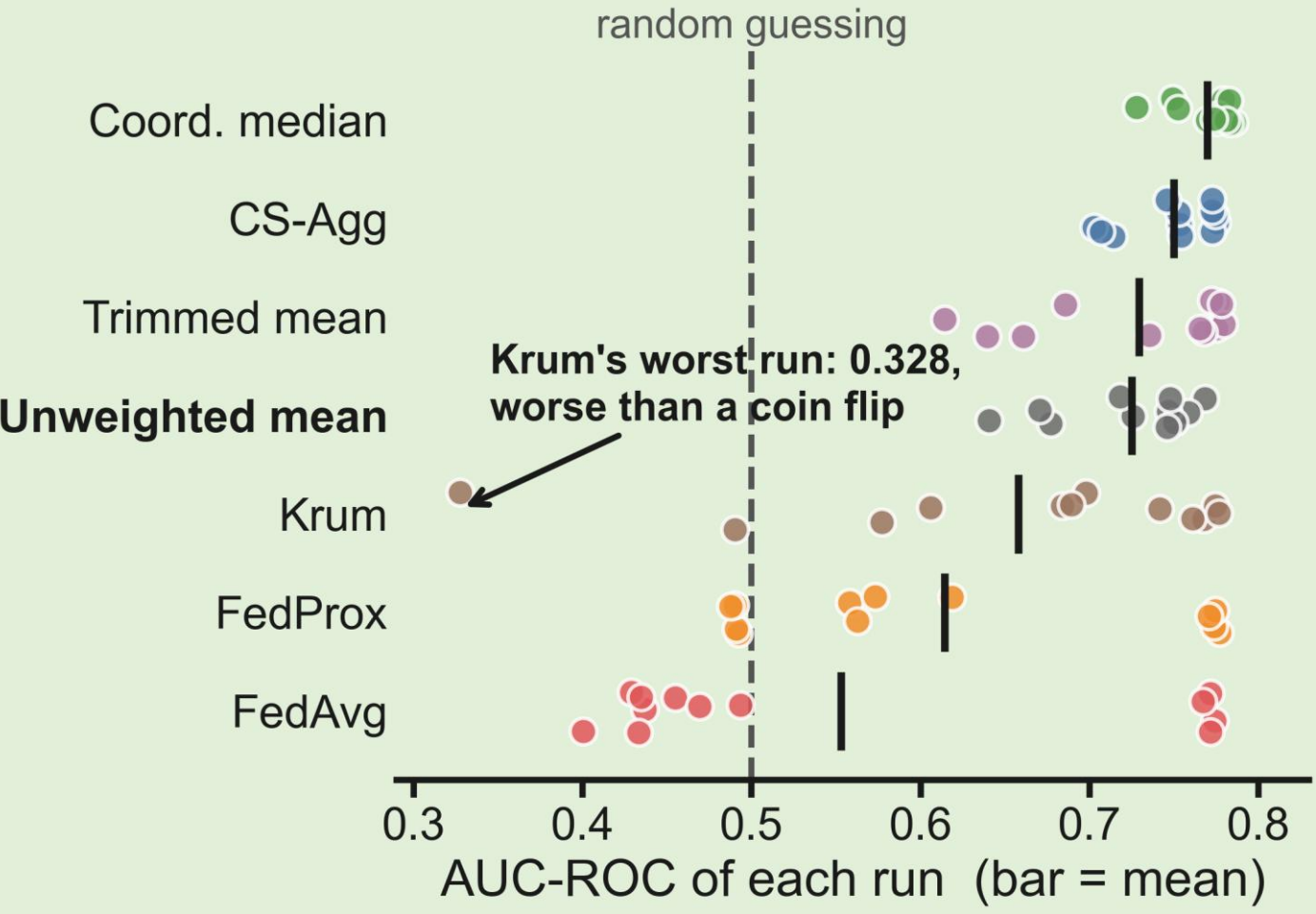


Merge rule	Mean AUC	Worst run	vs. FedAvg	vs. control	Holm p vs. control
FedAvg	0.7081	0.4002	—	-0.0584	0.0036
FedProx	0.7286	0.4874	+0.0205	-0.0379	0.0036
Krum	0.7439	0.3275	+0.0358	-0.0226	0.22 n.s.
Unweighted mean	0.7665	0.6405	+0.0584	—	control
Trimmed mean	0.7691	0.6143	+0.0610	+0.0026	0.0036
CS-Agg	0.7746	0.7025	+0.0665	+0.0081	0.22 n.s.
Coord. median	0.7828	0.7279	+0.0747	+0.0163	< 0.0001

36 runs per rule on BRFSS. The three robust rules that beat the control beat it by far less than the control beats FedAvg.

- The control alone recovers 79 to 98% of what the three strongest robust rules gain over FedAvg.
- Only coordinate-wise median clearly clears the control, and CS-Agg cannot be told apart from it at all.

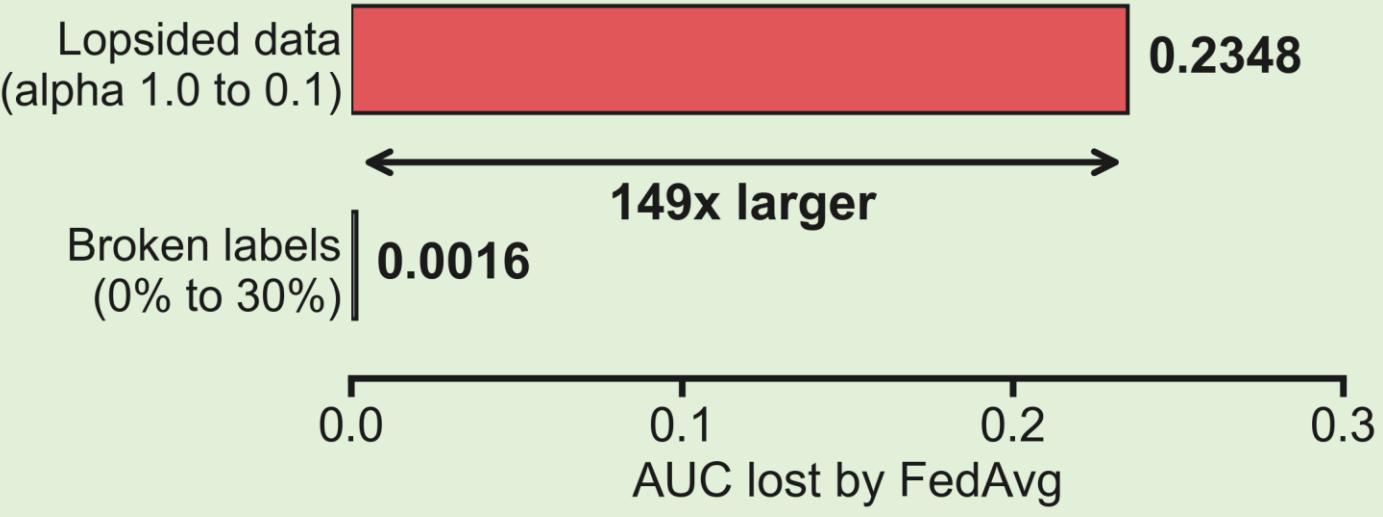
Krum is unreliable, not merely weaker



- Krum lands below the control and collapses to 0.328 AUC on one run, worse than a coin flip.

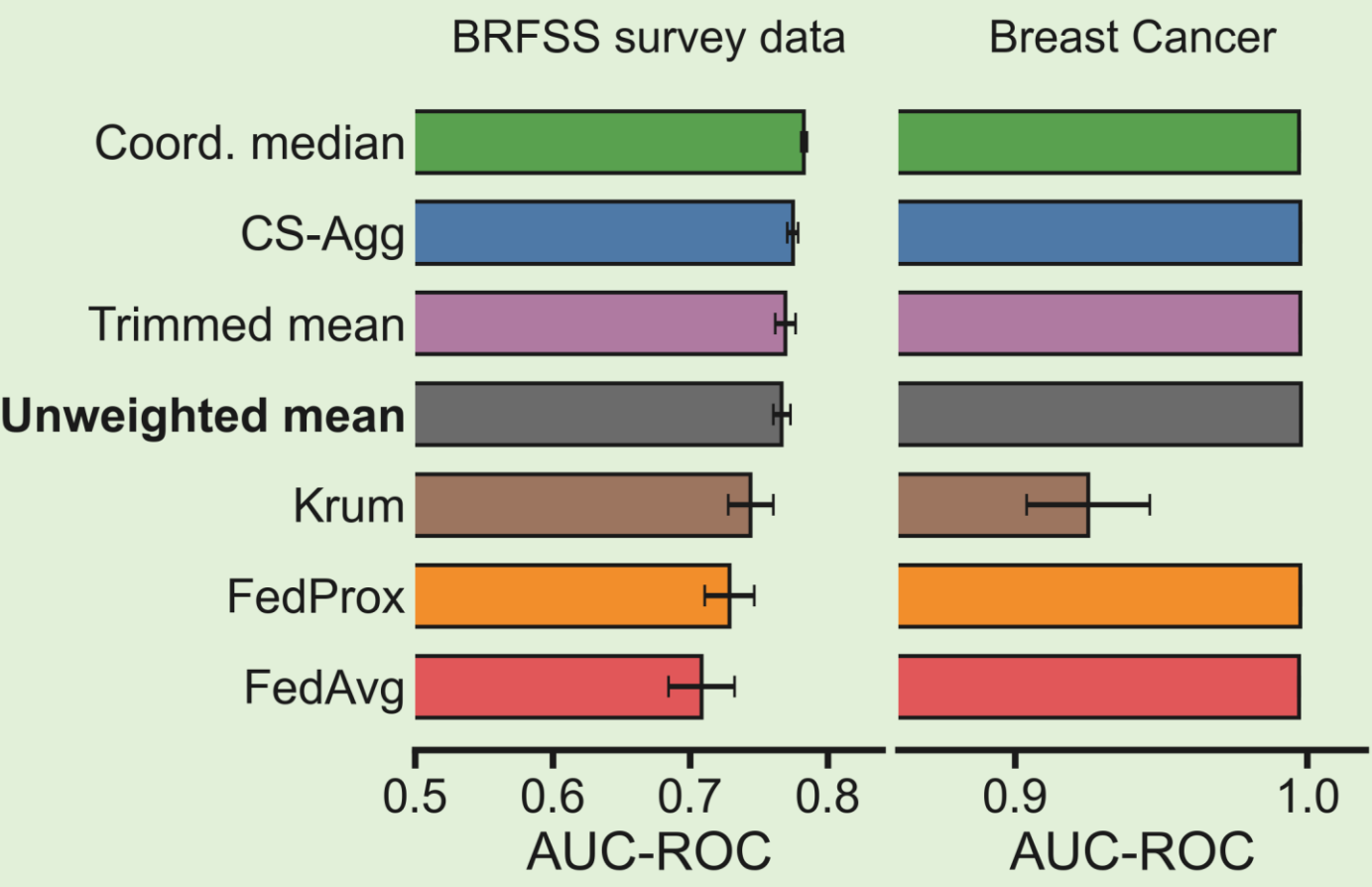
Results, continued

Lopsided data hurts far more than broken labels



- Moving from alpha = 1.0 to 0.1 costs FedAvg 0.2348 AUC, while 0% to 30% label noise costs it 0.0016.
- That 149x gap narrows to about 14x when sites track twelve health variables instead of four, but it never closes.

A clean benchmark would have hidden all of this



- On Breast Cancer every rule but Krum reaches 0.997 AUC or better, where no difference between them is visible.
- Messy, self-reported population health data is where these rules actually separate.

Future Directions

- Test adaptive Byzantine attackers, not only the label noise studied here.
- Replace the simulated Dirichlet split with data partitioned across real institutions.
- Check whether the weighting effect persists for deeper models and larger numbers of sites.

Conclusions

- Most of the advantage robust aggregators hold over FedAvg comes from how they weight sites, not from the robust statistic they are named for.
- Any robustness evaluation run on size-skewed data should include an unweighted-mean control.
- The control costs one line of code, and without it a reported gain cannot be credited to robustness.
- A rule's reputation is not evidence: Krum is widely cited and failed here, while coordinate-wise median is simple and held.

Code & Data

All code, the merged results and a script that re-checks every number on this poster are openly available.

github.com/kunalchevuri/federated-learning-health-robustness



Acknowledgements

- Survey data collected and published by the Centers for Disease Control and Prevention (BRFSS 2023).
- Carried out in the Computational Healthcare and Biotechnology Lab, University of North Texas.